7.2.6 Hollow Hemisphere

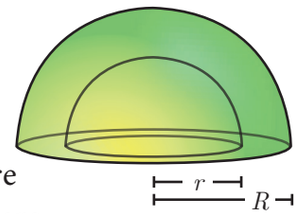


Fig. 7.17

Let the inner radius be r and outer radius be R,

then thickness $= R\text{-}r$

Therefore, C.S.A. $=$ Area of external hemisphere + Area of internal hemisphere

$= 2\pi R^2 + 2\pi r^2$

C.S.A. of a hollow hemisphere $= 2\pi(R^2 + r^2)$ sq. units

T.S.A. $=$ C.S.A. + Area of annulus region

$= 2\pi(R^2 + r^2) + \pi(R^2 - r^2)$

$= \pi\left[2R^2 + 2r^2 + R^2 - r^2\right]$

T.S.A. of a hollow hemisphere $= \pi(3R^2 + r^2)$ sq. units

Example 7.8 Find the diameter of a sphere whose surface area is 154 m^2.

Solution Let r be the radius of the sphere.

Given that, surface area of sphere $= 154$ m^2

$$4\pi r^2 = 154$$

$$4 \times \frac{22}{7} \times r^2 = 154$$

$$\Rightarrow \quad r^2 = 154 \times \frac{1}{4} \times \frac{7}{22}$$

$$r^2 = \frac{49}{4} \text{ we get } r = \frac{7}{2}$$

Therefore, diameter is 7 m

Activity 3

Using a globe, list any two countries in the northern and southern hemispheres.

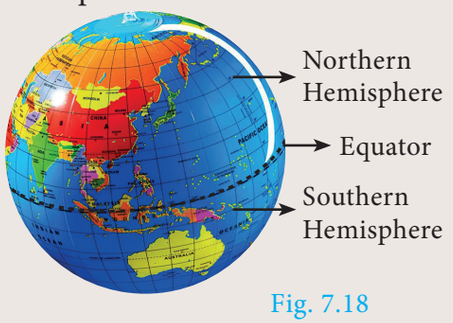


Fig. 7.18

Example 7.9 The radius of a spherical balloon increases from 12 cm to 16 cm as air being pumped into it. Find the ratio of the surface area of the balloons in the two cases.

Solution Let r_1 and r_2 be the radii of the balloons.

Given that, $\frac{r_1}{r_2} = \frac{12}{16} = \frac{3}{4}$

Now, ratio of C.S.A. of balloons $= \frac{4\pi r_1^2}{4\pi r_2^2} = \frac{r_1^2}{r_2^2} = \left(\frac{r_1}{r_2}\right)^2 = \left(\frac{3}{4}\right)^2 = \frac{9}{16}$

Therefore, ratio of C.S.A. of balloons is 9:16.

Thinking Corner

1. Find the value of the radius of a sphere whose surface area is 36π sq. units.
2. How many great circles can a sphere have?
3. Find the surface area of the earth whose diameter is 12756 kms.

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